

Contributors Dashboard

Project Documentation

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1. Introduction

The Contributors Dashboard is a web-based application built to bring information such as shops, cooperatives, schools, health posts, government offices, farms, and many other establishments together in one place. It combines a spreadsheet of contributor records with an interactive map of the municipality, so that municipal staff can filter, search, and visualise contributor data by category, by ward, and by points earned. The dashboard is designed to be simple enough for non-technical staff to use day-to-day, while being built on a foundation that can later be connected to a live municipal database.

This document explains the purpose of the dashboard, the approach and tools used to build it, and the outcome achieved in the current version. We can understand what the dashboard does, how it was built, and how it can be extended in the future.

2. Objective

The main objective of this project is to design and develop a Contributors Dashboard for the municipality that makes contributor data easy to view, filter, and understand. The specific objectives are:

- **Centralise contributor records:** bring together contributor name, category, address, ward, geographic location, and points earned into a single, organised dataset.
- **Visualise contributors on a map:** show each contributor as a marker on an interactive map of a municipality, placed within the correct ward boundary.
- **Enable category-based filtering:** allow staff to filter contributors by one or more categories (such as Agriculture, Education, Health, Tourism, and others) to focus on a specific sector.
- **Enable ward-based filtering:** allow staff to select a specific ward and instantly see only the contributors and boundary belonging to that ward.
- **Track and rank points:** calculate total points earned per contributor and per category, and present a leaderboard of the top-performing contributors.
- **Support search and quick review:** provide a searchable, sortable table view of contributor records alongside the map, so that individual records can be found quickly.
- **Reflect municipal identity:** display the official emblem of the municipality on the dashboard so that it is clearly recognised as an official municipal tool.
- **Prepare for future database integration:** build the dashboard in a way that its current spreadsheet-based data source can later be replaced with a live database connection, without redesigning the dashboard itself.

3. Methodology

The dashboard was developed by combining four building blocks: a contributor dataset, a geographic boundary file for the wards, the official municipal logo, and a web-based mapping interface that ties everything together. Each of these is explained below, along with the tools and reasoning used at each step.

3.1 Overall Approach

The dashboard was built as a self-contained web application using standard web technologies such as HTML for structure, CSS for styling, and JavaScript for interactivity which is combined with the open-source Leaflet mapping library. This approach was chosen because it does not require any specialized software to run and the dashboard opens directly in a web browser and can be hosted on any standard web server or municipal computer without additional installation.

The design follows a single page layout: a header showing the municipality's identity and summary statistics, a toolbar for filtering by category, a map on one side, and a searchable table of records on the other. This layout was chosen so that staff can see the location of a contributor and its record details side by side, without switching between separate tools.

3.2 Municipality Branding (Logo Integration)

To make sure the dashboard is clearly identified as an official Municipality tool, the official emblem of the municipality has been added to the header section of the dashboard, alongside the municipality's name in both Nepali and English. For the demonstration Dhulikhel Municipality logo is taken as a reference along with its name. This gives the dashboard an official, trustworthy appearance and distinguishes it from a generic or third-party tool. The logo file is stored alongside the other dashboard files and is loaded automatically whenever the dashboard is opened.

3.3 Ward Boundary Mapping (GeoJSON)

One of the core requirements was to show contributors within the correct ward boundaries of the municipality, rather than on a plain, unmarked map. To achieve this, the boundaries of all wards were prepared as a GeoJSON file; a standard, widely-used geographic data format that stores the outline (polygon shape) of each ward along with its ward number as a property.

This GeoJSON file is loaded into the dashboard and drawn on top of the base map using the Leaflet mapping library. Each ward boundary is rendered as a distinct shape that can be clicked to select that ward. When a ward is selected:

- The selected ward's boundary is highlighted with a darker outline and a stronger fill color, while all other wards remain in their default, lighter style.
- The map automatically zooms and centers on the selected ward for a clearer view.
- A ward selection panel lists all wards as clickable chips, so a ward can be chosen either directly on the map or from this list.

Using GeoJSON for the ward boundaries means that the boundary data is kept independent of the contributor data. If ward boundaries are redrawn or updated by the municipality's survey or planning office in the future, only the GeoJSON file needs to be replaced — the rest of the dashboard continues to work without any change to its code.

3.4 Contributor Data Management (Excel-based Source)

The contributor list — including each contributor's name, the responsible contact person, category, address, ward number, points earned, and geographic coordinates (latitude and longitude) is maintained as an Excel spreadsheet. This format was chosen for the current version because it is easy to update without technical assistance, and simple to review and is taken as a reference only.

The fields currently maintained for each contributor record are:

Field	Description
Name	Name of the contributing establishment (e.g., shop, cooperative, school, office).
Contributor	The individual associated with or responsible for the record.
Category	The sector the contributor belongs to (e.g., Agriculture, Health, Tourism).
Address	Ward-level address of the contributor.
Ward	Ward number, used for ward-based filtering and map boundary matching.
Points	Total points earned by the contributor through municipal participation.
Latitude / Longitude	Geographic coordinates used to place the contributor as a marker on the map.

For the dashboard to read this information, the Excel data is converted into a structured data file that the web page can load directly. This conversion step keeps the dashboard fast and independent of any spreadsheet software while the underlying source of truth remains the Excel file that municipal staff maintain.

3.5 Category and Points System

Contributors are grouped into a fixed set of categories defined in coordination with the municipality (for example: Agriculture, Commercial, Communication, Education, Entertainment, Food and Drinks, Finance, Fuel, Games, Government, Health, Industry, Infrastructure, Lifestyle, Natural, Organization, Place, Security, Shopping, Vehicle Service, Services, Stay and Travel, and Tourism). Each category has been assigned its own icon and color so that markers on the map can be recognized at a glance without needing to open each one individually.

Points earned by each contributor are recorded against their category. The dashboard automatically calculates, for every category, the number of contributors and the total points earned, and it calculates an overall leaderboard by summing points per contributor across all categories combined. This removes the need for manual counting or calculation by staff.

3.6 Map Visualization and Interaction (Leaflet)

The interactive map is built using Leaflet, an open-source JavaScript mapping library, with Galli Maps as the underlying base map layer. This combination was selected as per our requirement.

On top of the base map, the dashboard adds the following layers and interactions:

- **Contributor markers:** each visible contributor is plotted as a colored, category-specific pin at its recorded latitude and longitude.
- **Ward boundaries:** drawn from the GeoJSON file, clickable for the ward-based selection as described in Section 3.3.
- **Category filter:** a panel allowing one or more categories to be selected or cleared, with the map and table updating instantly to match the selection.
- **Legend:** shown automatically whenever categories are active, listing the color and icon used for each selected category.
- **Summary statistics:** a strip in the header showing the number of active categories, number of contributors shown, and total points shown, recalculated live as filters change.

3.7 Contributor Records Table and Search

Alongside the map, a table lists every contributor matching the current filter, showing name, contributor, category, address, and points. A search box above the table allows to search across category name, contributor person, and address of the category at the same time, narrowing the table (and the map) to matching results instantly. This gives staff two complementary ways to explore the data — visually on the map, or as a precise, searchable list.

3.8 Leaderboard and Ward Panel

A "Top 10" leaderboard panel ranks contributors by total points earned across all categories, giving the municipality a quick view of its most active contributors. A separate ward panel lists all wards and highlights the currently selected ward, and includes a "Clear" option to remove the ward filter and return to the full municipal view. A "Refresh" button resets all filters, searches, and selections back to the default view of the full municipality.

For the overall code [Click Here](#)

3.9 Technology Stack Used

Component	Technology / Format Used
Structure & content	HTML5
Styling & layout	CSS3 (custom stylesheet)
Interactivity & logic	JavaScript (client-side)
Mapping engine	Leaflet.js (open-source)
Base map tiles	Galli Maps
Ward boundaries	GeoJSON
Contributor records	Microsoft Excel (source), converted to a structured data file for the dashboard
Municipal branding	Official municipality emblem (image file)

4. Results

The current version of the Contributors Dashboard has been completed and is functional. The following outcomes have been achieved:

- **A unified, branded dashboard:** The dashboard opens with the official Dhulikhel Municipality logo (taken as a reference municipality), name (in Nepali and English), and a live summary strip showing active categories, contributors shown, and total points giving an immediate, official looking overview of the municipality's contributor base.
- **Full ward coverage on the map:** all municipal wards are displayed as boundaries drawn from the GeoJSON file, each clickable and selectable, with the map automatically zooming to the selected ward.
- **Category-based filtering that works instantly:** combination of the 23 defined categories can be selected, and both the map marker and the contributor table update immediately, along with a legend and per-category counts.
- **A working points and leaderboard system:** total points are calculated automatically per category and per contributor, and the Top 10 leaderboard correctly ranks contributors across the full dataset.
- **A searchable records table:** the table view lists every contributor matching the active filters, with a working search box that filters by name, contributor, and address in real time.
- **A usable, adaptable layout:** the map and table panels can be resized by dragging a divider between them, and a "Refresh" control resets all filters and view settings back to the default municipal-wide view in one click.
- **A clear path to live data:** because contributor and ward data are loaded from clearly separated files (the Excel-based contributor list and the GeoJSON ward file), the dashboard is ready to be connected to a GeoServer-backed spatial database, without needing to redesign the map, table, filters, or leaderboard.

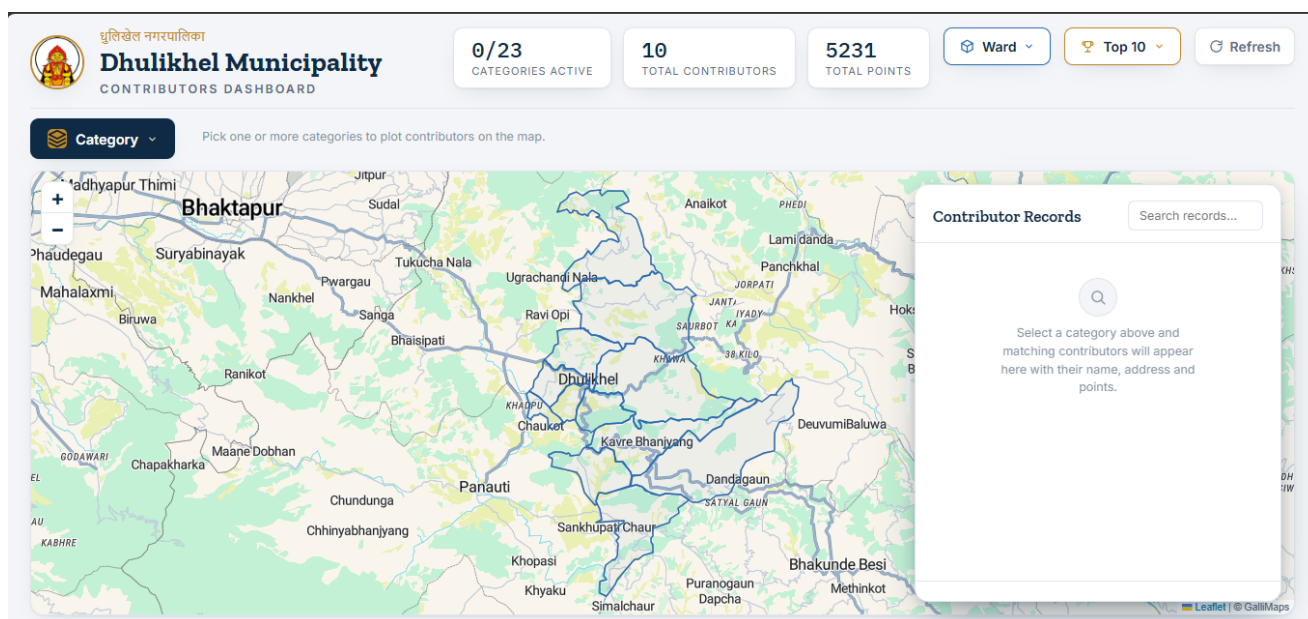


Figure 1: Dashboard Interface

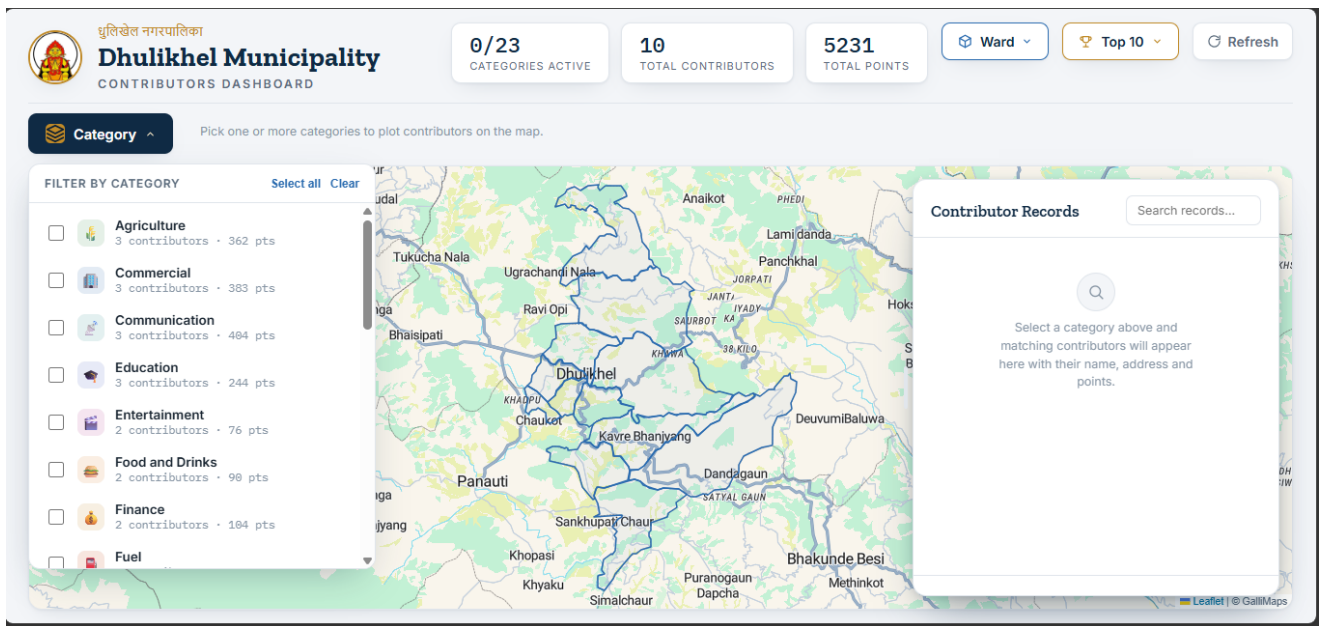


Figure 2: Screenshot showing the category lists.

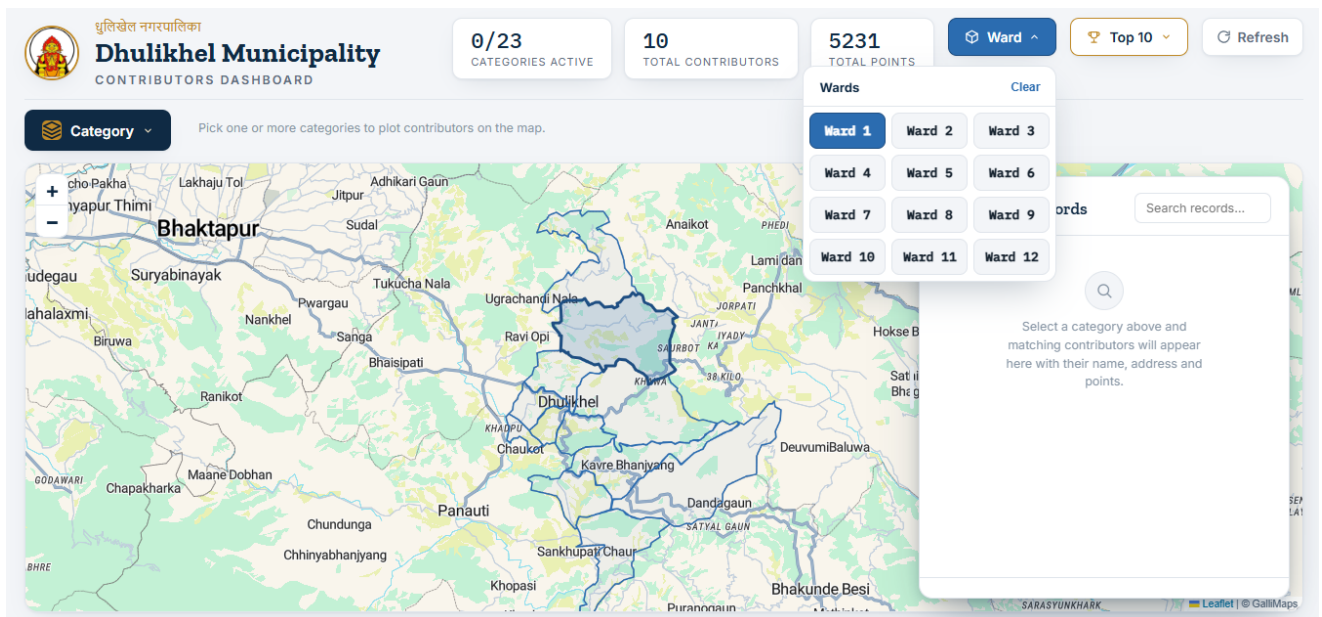


Figure 3: Screenshot showing the ward selection option

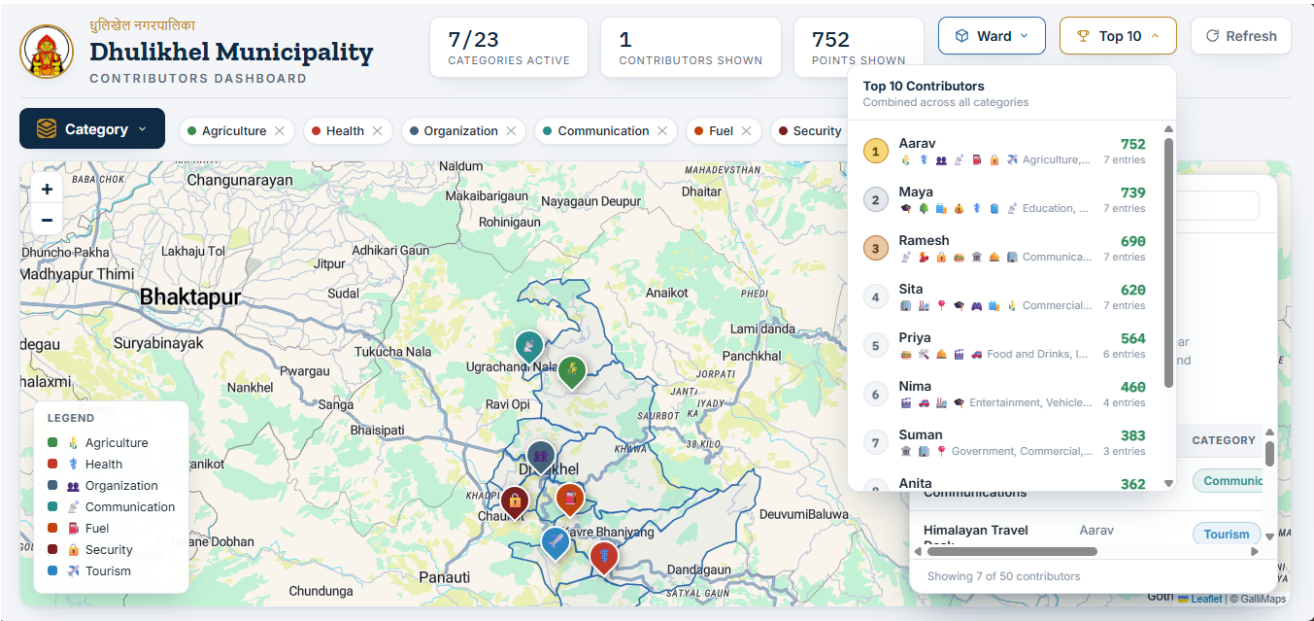


Figure 4: Screenshot showing the Top Contributors list

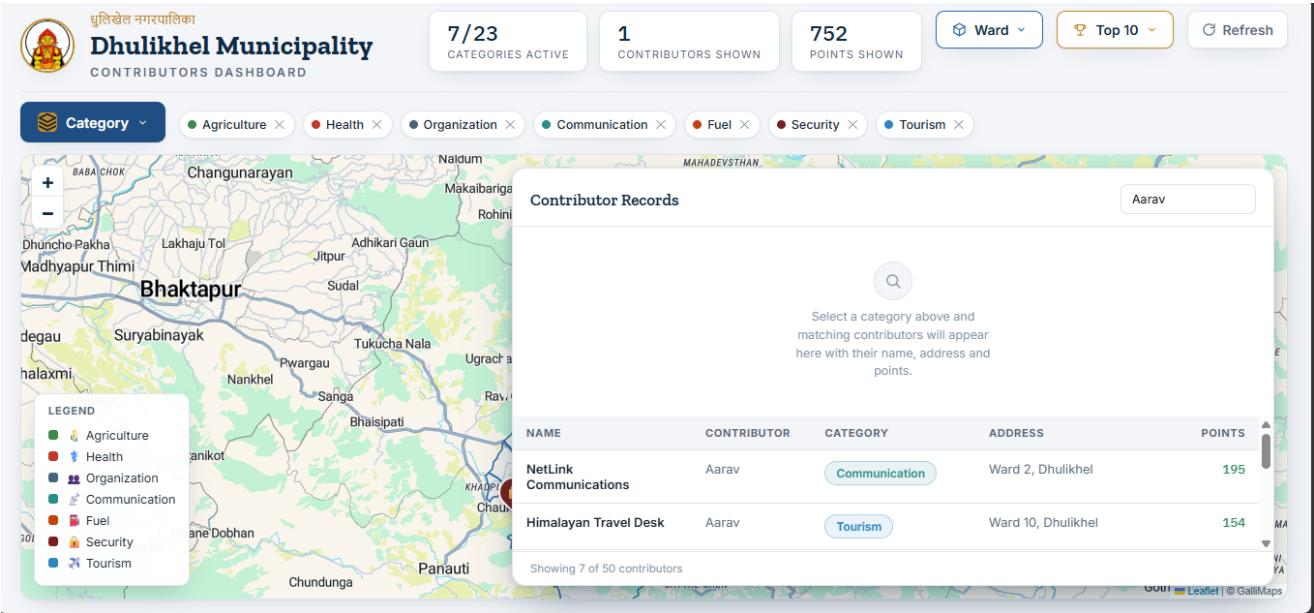


Figure 5: Screenshot illustrating contributor points awarded for completed contributions

5. Future Implementation

Overall, the dashboard successfully meets its objective of giving any Municipality a single, visual, and interactive tool to monitor contributor participation by category, ward, and points; while remaining straightforward for non-technical staff to use and easy to extend as the municipality's data infrastructure grows.

The current version of the dashboard uses an Excel spreadsheet as the source of contributor information and a static GeoJSON file for the ward boundaries. These formats were intentionally selected to enable rapid deployment, allowing the municipality to begin using and evaluating the dashboard without requiring a dedicated database infrastructure. As the system grows and more frequent data updates become necessary, the dashboard can be enhanced to support a centralized spatial database and GeoServer-based architecture. A recommended future implementation approach is outlined below:

Migrate the existing data

The contributor records currently stored in Excel and the ward boundary GeoJSON file can be migrated to a spatial database such as PostgreSQL/PostGIS. This will provide centralized storage for both attribute and spatial data.

Configure GeoServer

GeoServer can then be connected to the spatial database to publish contributor locations and ward boundaries as standard web mapping services (e.g., WMS/WFS), making the data available for web-based applications.

Update the dashboard data source

The dashboard can be modified to retrieve contributor and ward data directly from GeoServer services instead of reading local Excel and GeoJSON files. Since the dashboard has been designed with a clear separation between the data source and the visualization components, this change would primarily involve updating the data retrieval mechanism while keeping the user interface largely unchanged.

Enable live data updates

Once the dashboard is connected to the spatial database through GeoServer, any changes made to the central database such as adding new contributors, updating contribution values, or modifying ward boundaries—would be reflected automatically in the dashboard without requiring manual updates to Excel or GeoJSON files.

This phased implementation approach allows the municipality to begin using the dashboard immediately with simple file-based data while providing a clear pathway toward a scalable, centrally managed, and real-time spatial information system as future infrastructure becomes available.